

Hydrophobic modules and their disruption: signatures of purifying selection in human populations

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ABSTRACT

Hydrophobic residues cluster into contiguous runs that behave like modular units of protein structure. Using 2,504 genomes from the 1000 Genomes Project, we asked whether coding variants that redraw a blob boundary ($H \leftrightarrow P$) are held at lower diversity than amino-acid changes that leave the module intact ($H \rightarrow H$, $P \rightarrow P$). Genome-wide, $H \leftrightarrow P$ variants show lower per-transcript heterozygosity ratio R in both thresholds across five global populations (CEU, CHB, CLM, ESN and YRI). Gene Ontology (GO) analysis reveals that transcripts with reduced diversity $H \leftrightarrow P$ sites are enriched for essential physiological functions such as ion channel activity and transmembrane transport, whereas sites with excess diversity cluster in olfactory and sensory receptors.

RESEARCH QUESTIONS

- Are contiguous hydrophobic modules (h-blobs) under purifying selection in humans?
- Is diversity reduced at blob-disrupting sites, beyond what amino-acid change or secondary structure already explain?

APPROACH

Blobulator clusters a protein sequence into hydrophobic (h) and polar (p) blobs, assigning each residue to one of them. A variant either leaves the residue's blob state unchanged ($H \rightarrow H$, $P \rightarrow P$) or flips it ($H \leftrightarrow P$) and synonymous and nonsynonymous variants serve as the neutral and amino-acid-change baselines. Our approach here is to measure the average heterozygosity of variants and genes across different classes of genetic variation in a panel of five populations (CEU, CHB, CLM, ESN, and YRI). We group variants in three ways: whether the substitution changes the blob state, whether the residue sits in an h or p blob in the reference sequence, and its secondary structure in the reference sequence.

Blobulate the human reference proteome and known variants. We blobulated the reference allelic state of every amino acid in every protein and the alternate allelic state of each of 943,260 coding SNPs. Every individual in the 1000 Genomes Project ($n = 2,504$) is then given a blob "genotype" at each site.

Record secondary structure states. DSSP was used to categorize the reference state of every amino acid in every transcript as helix, sheet or coil [1,2].

Calculate mean and relative heterozygosity in each population. We compute mean heterozygosity for each class of variants and relative heterozygosity R (box, right) per transcript, giving distribution of R across transcripts for each class and population. Lower-than-expected heterozygosity at a locus can indicate either purifying or positive selection.

Calculate Tajima's D over variant classes. D measures the excess of rare variants; more negative values indicate stronger purifying selection. Computed with [3].

Test for pathway enrichment. Genes were ranked by pooled R , keeping those with ≥ 6 segregating variants. GO enrichment analysis (BP, MF and CC) was conducted using g:Profiler [4]. Tested with Benjamini-Hochberg correction against background of genes meeting the same threshold.

Relative heterozygosity

Relative heterozygosity R is the mean heterozygosity of variants of a given class within a transcript, in a given population, divided by the mean heterozygosity of all segregating variants in that transcript. Normalizing within transcripts allows us to compare classes across genes with markedly different levels of genetic diversity.

$$P_{ic} \text{ variant allele frequency at site } i \text{ in population } c$$

$$H_{ic} = 2P_{ic}(1 - P_{ic}) \text{ heterozygosity of variant } i \text{ in population } c$$

$$\bar{H}_{jc,cat} \text{ mean } H_{ic} \text{ over one category in protein } j, \text{ population } c$$

$$\bar{H}_{jc,tot} \text{ mean } H_{ic} \text{ over all variants in protein } j, \text{ population } c$$

$$R = \frac{\bar{H}_{jc,cat}}{\bar{H}_{jc,tot}}$$

$R < 1$ = reduced diversity · $R > 1$ = excess diversity

BACKGROUND

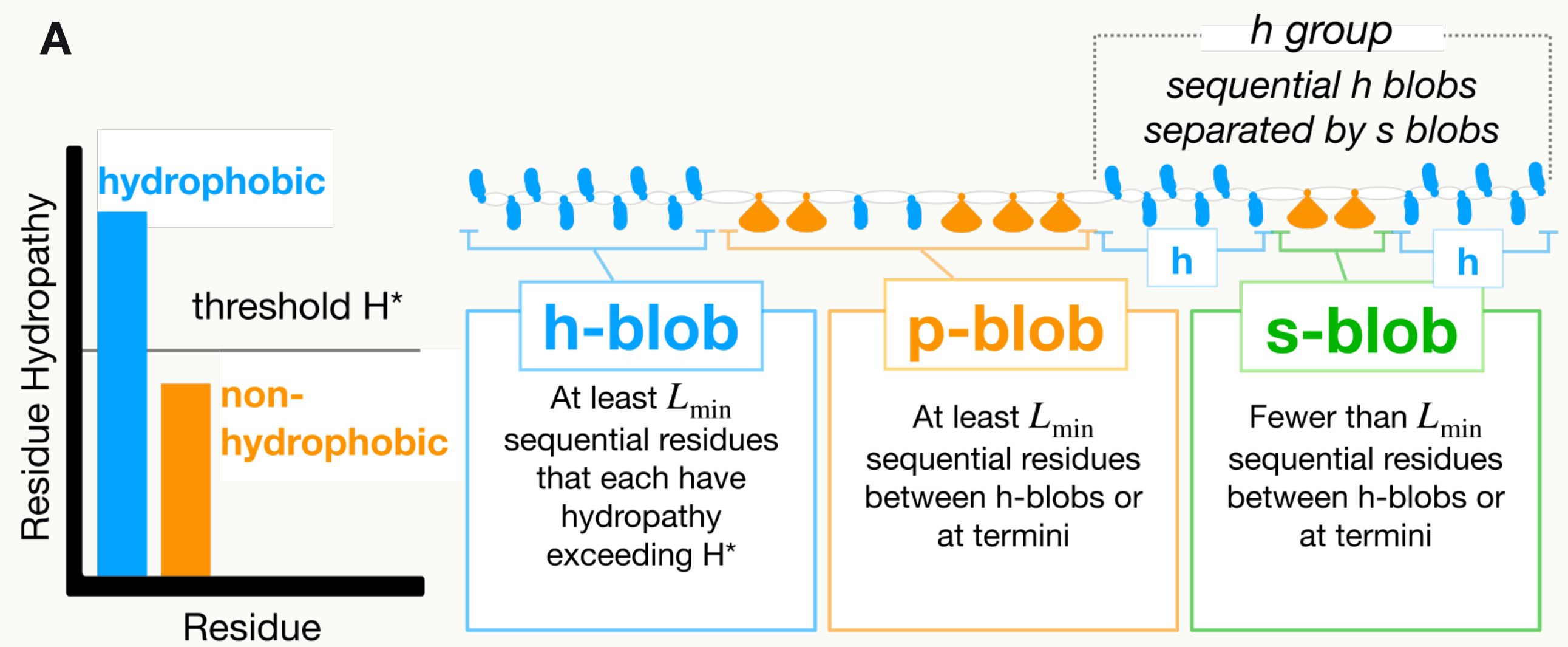


Figure 1. Global genetic variation and the measure of contiguous hydrophobicity. **(A)** Residues are classified as hydrophobic or non-hydrophobic based on Kyte-Doolittle hydrophathy [5] falling above or below the user-defined threshold. Next, the clustering step acts on the digitized sequence with the user-provided minimum blob size (adapted from [6]). **(B)** The map shows patterns of genetic polymorphism in the 1000 Genomes Project. The pie charts depict variants shared across all continents, specific to a continent, or specific to a population (adapted from [7]).

RESULTS 1 Is diversity reduced at blob-disrupting sites?

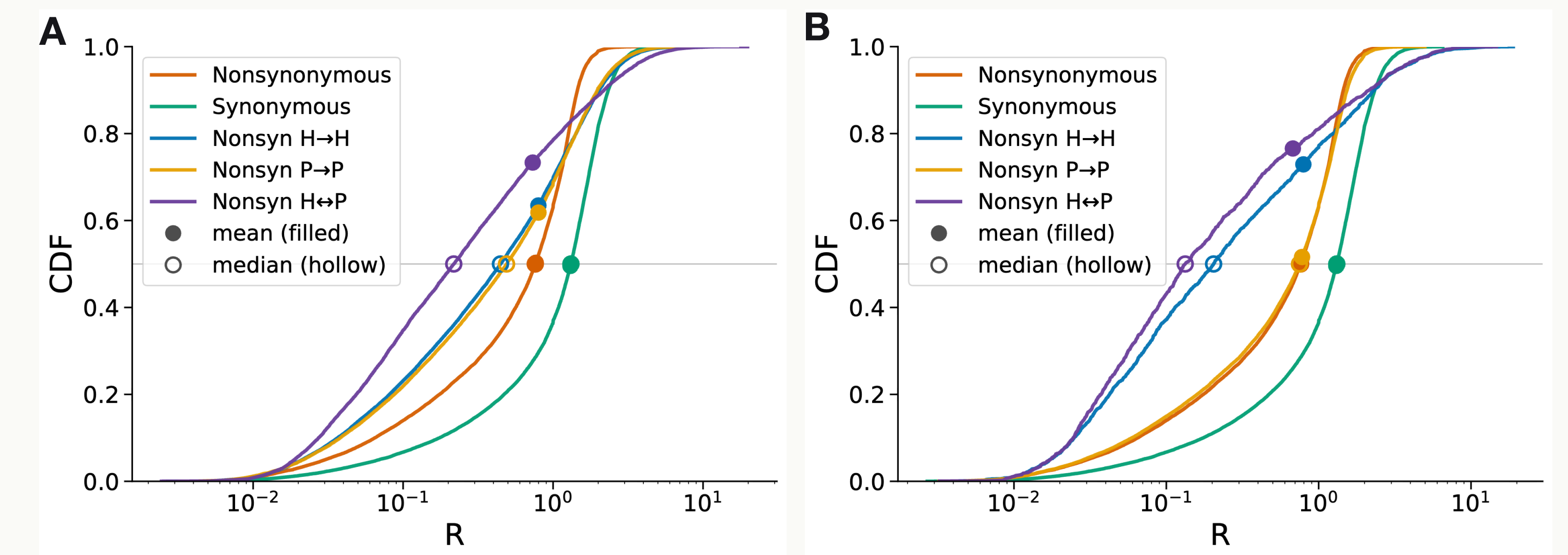


Figure 2. Cumulative distribution functions (CDFs) of per-transcript relative heterozygosity R , for five variant classes (colors) at **(A)** $H^* = 0.4$, $L = 4$ and **(B)** $H^* = 0.6$, $L = 5$.

RESULTS 2 Does blob state add anything beyond secondary structure?

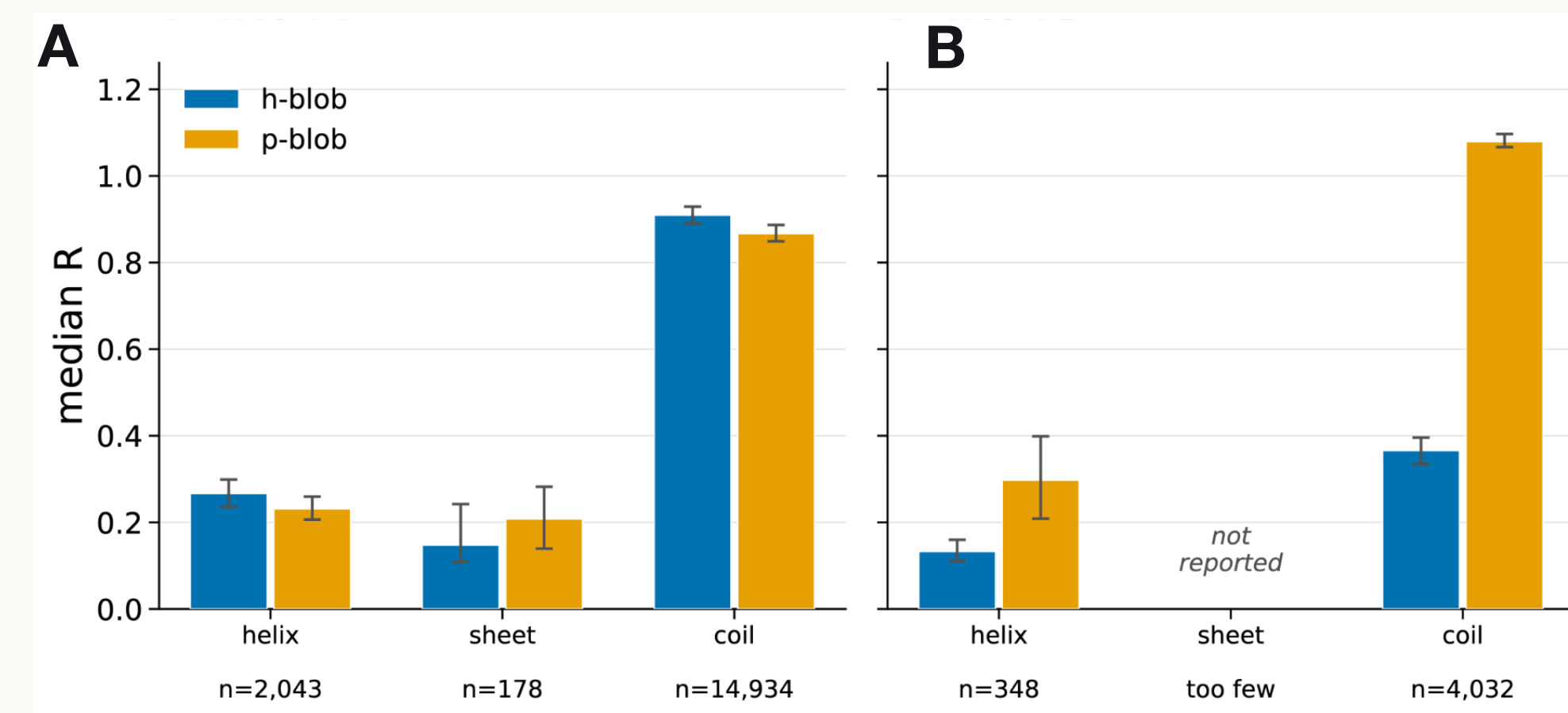


Figure 3. R for h-blob vs p-blob sites by secondary structure, pooled across populations, at **(A)** $H^* = 0.4$, $L = 4$ and **(B)** $H^* = 0.6$, $L = 5$. Helix and sheet sites are more constrained than coil sites. The differences in h/p are inconsistent, small in (A), large in (B) coil. A within-transcript ANOVA, where each gene is its own control, confirms that structure shows $\sim 5\times$ more variance in R than blob state (ω^2 0.021% vs 0.004% at $H^* = 0.4$), blob state explains nothing at $H^* = 0.6$ (ω^2 $\sim 0.001\%$, $p = 0.999$). Error bars are 95% CIs; n is the number of transcripts in each bar.

RESULTS 3 Which functions are enriched among genes with most and least h-blob diversity?

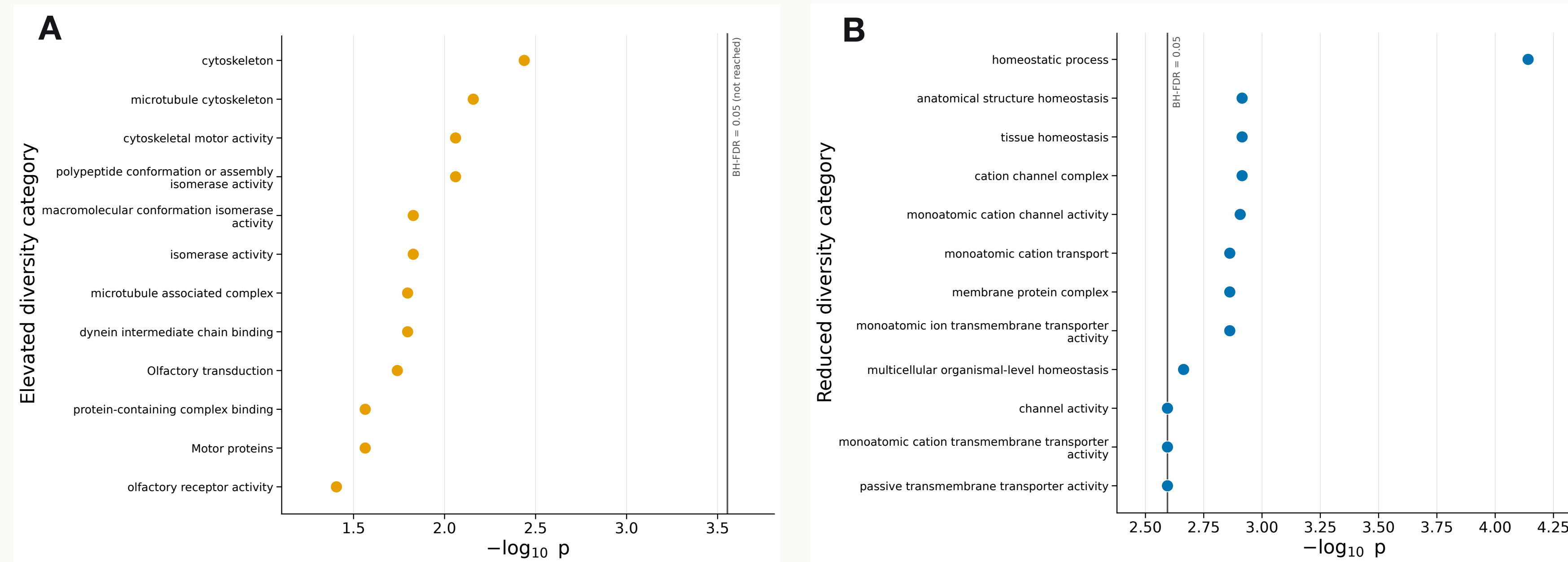


Figure 4. GO enrichment for the 35 genes with the most elevated diversity versus the 35 with the most reduced diversity, drawn from 179 genes with ≥ 6 segregating $H \leftrightarrow P$ variants at $H^* = 0.6$, $L = 5$, tested against a background of all 179 genes. **(A)** No term reached FDR < 0.05 among the elevated-diversity genes. **(B)** The reduced-diversity genes were enriched for ion channel activity, transmembrane transport activity, membrane protein complex, and anatomical structure, functional categories in which strong purifying selection is expected.

RESULTS 4 Do allele frequencies depart from neutrality in different blob categories?

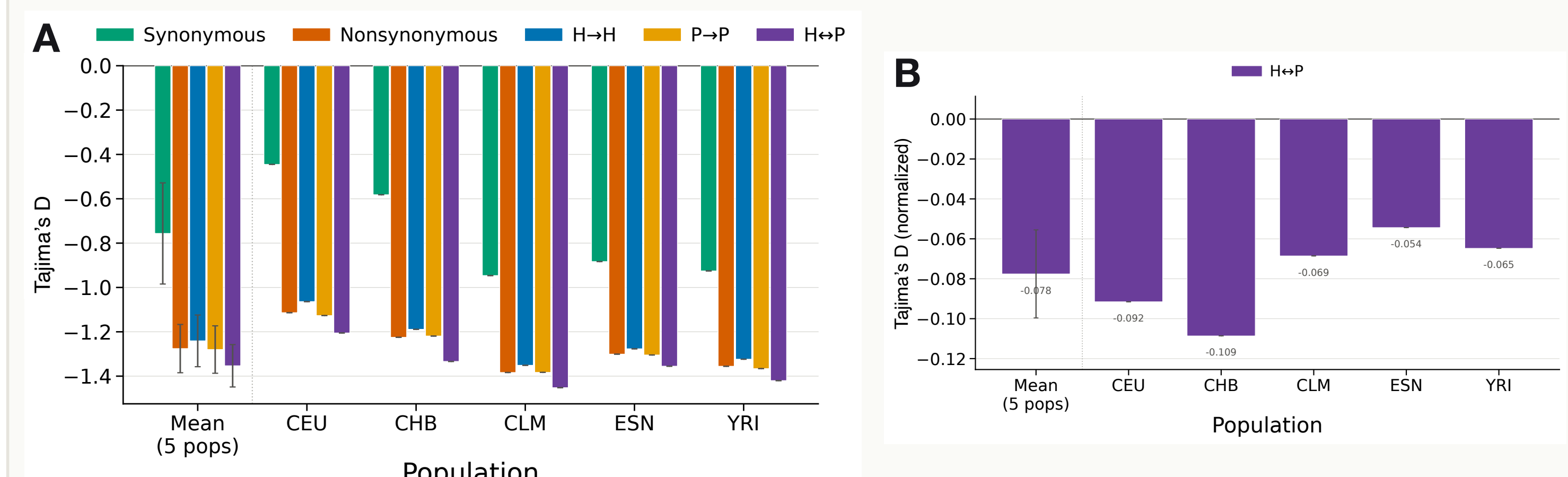


Figure 5. Tajima's D [8] by variant category in CEU, CHB, CLM, ESN and YRI at $H^* = 0.4$, $L = 4$. **(A)** Under this threshold, variants that disrupt H-blob defining category have the lowest D value across populations signifying a consistent pattern across demographic history and the CLM has the lowest Tajima's D for all variant categories. **(B)** Difference in D between $H \leftrightarrow P$ and all nonsynonymous variants in the same population, $\Delta D = D(H \leftrightarrow P) - D(\text{nonsynonymous})$. Because nonsynonymous variants include $H \leftrightarrow P$, this comparison is conservative.

CONCLUSIONS

- 1 Variants that disrupt hydrophobic module ($H \leftrightarrow P$) have lower genetic diversity than variants that leave it intact.
- 2 At a lower hydrophobicity threshold, secondary structure explains diversity better than the blob state does. At the stricter threshold the two are more comparable.
- 3 Among genes with at least six h-blob-altering variants at the high threshold, the 35 with lowest diversity in those variants are enriched for ion channel and transmembrane transport functions.
- 4 $H \leftrightarrow P$ variants have the most negative Tajima's D in each population, a pattern expected in purifying selection.

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